STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 3

Description

3. Demonstration of classification rule process on dataset employee.arff using j48 algorithm

Aim: This experiment illustrates the use of j-48 classifier in weka. The sample data set used in this experiment is "student" data available at arff format. This document assumes that appropriate data pre processing has been performed.
Steps involved in this experiment:
Step-1: We begin the experiment by loading the data (student.arff)into weka.
Step2: Next we select the "classify" tab and click "choose" button t o select the "j48"classifier.
Step3: Now we specify the various parameters. These can be specified by clicking in the text box to the right of the chose button. In this example, we accept the default values. The default version does perform some pruning but does not perform error pruning.
Step4: Under the "text" options in the main panel. We select the 10-fold cross validation as our evaluation approach. Since we don't have separate evaluation data set, this is necessary to get a reasonable idea of accuracy of generated model.
Step-5: We now click "start" to generate the model .the Ascii version of the tree as well as evaluation statistic will appear in the right panel when the model construction is complete.
Step-6: Note that the classification accuracy of model is about 69%.this indicates that we may find more work. (Either in preprocessing or in selecting current parameters for the classification)
Step-7: Now weka also lets us a view a graphical version of the classification tree. This can be done by right clicking the last result set and selecting "visualize tree" from the pop-up menu.
Step-8: We will use our model to classify the new instances.
Step-9: In the main panel under "text" options click the "supplied test set" radio button and then click the "set" button. This wills pop-up a window which will allow you to open the file containing test instances.

Description:
Classification is a data mining function that assigns items in a collection to target categories or classes. The goal of classification is to accurately predict the target class for each case in the data. For example, a classification model could be used to identify loan applicants as low, medium, or high credit risks.
Entropy characterizes the (im) purity of an arbitrary collection of examples
Information Gain is the expected reduction in entropy caused by partitioning the examples according to a given attribute
If we have a set with k different values in it, we can calculate the entropy as follows:

Procedure and Overvation:

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Kappa statistic is a measure of how closely the instances classified by the machine learning classifier matched the data labeled as ground truth, controlling for the accuracy of a random classifier as measured by the expected accuracy.
Decision tree builds classification or regression models in the form of a tree structure. It breaks down a dataset into smaller and smaller subsets while at the same time an associated decision tree is incrementally developed. The final result is a tree with decision nodes and leaf nodes.

Classifier Accuracy Measures
The confusion matrix is a useful tool for analyzing how well your classifier can recognize tuples of different classes.

F measure (F1 score or F score) is a measure of a test's accuracy and is defined as the weighted harmonic mean of the precision and recall of the test.

TP Rate: rate of true positives (instances correctly classified as a given class)

FPRate: rate of false positives (instances falsely classified as a given class)

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Precision is the fraction of retrieved data that are relevant to the query:

Recall is the fraction of the relevant data that are successfully retrieved.

Cross-validation: In k-fold cross-validation, the initial data are randomly partitioned
into k mutually exclusive subsets or “folds,” D1, D2: Dk, each of approximately equal
size.

Dataset employee.arff using j48 algorithm


@relation employee

@attribute age {25,27,28,29,30,35,48}

@attribute salary {10k,15k,17k,20k,25k,30k,34k,35k,32k}

@attribute performance {good,avg,poor}

@data


%

25,10k,poor

27,15k,poor

27,17k,poor

28,17k,poor

29,20k,avg

30,25k,avg

29,20k,avg

30,20k,avg

35,32k,good

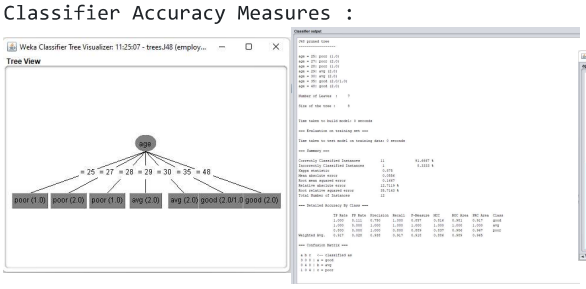
48,34k,good

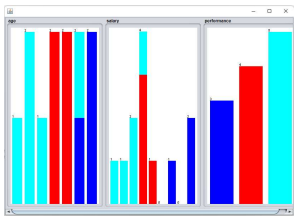
48,32k,good

35,20k,poor

%
```

Classifier Accuracy Measures :





VIVA RESPONSES

Q) ETL stands for _____

Extract, transform, and load

Q) Identify among the following for which system of data warehousing is mostly used.

Reporting and data analysis

Q) Small logical units where data warehouses hold large amounts of data is known as ____

Data miners

Q) Identify the main characteristic of OLTP.

Provides advanced database support

Q) Under which of the following does the pattern evaluation issue fall?

Mining methodology and user interaction issue.